Open problems:
Classical uncertainty principle

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`). Items marked [BLOCKS VERIFICATION] stand in the way of checking a main claim of the brief as the brief states it: a *formalization gap* can be closed in Lean alone, while a *brief issue* needs the brief to be corrected or made precise first.

No item below blocks the verification of a main claim of the brief.

From the brief

1. **The quantum case.** The conclusion says that "in general, the uncertainty principle is a consequence of the entropic lower bound", since pure quantum states have zero entropy. The classical bound $\hbar/e \approx 0.37\,\hbar$ is weaker than Heisenberg's $\hbar/2$, and no entropic derivation of the quantum bound is given. Whether, and in what form, the quantum principle follows from the zero entropy of pure states is open. (Brief, Sec. 3.)

From the formalization

1. None found.